PRONOIA: Event-Driven Financial Reasoning with Market-Verifiable Rewards

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ABSTRACT

Event-driven equity trading is shifting from speed alpha to cognitive alpha: reading an announcement's implications more deeply than the market. Two gaps block progress: traditional event study methods are post-hoc descriptive and cannot form forward-looking quantifiable predictions; and LLM-enhanced methods can read announcements but universally rely on LLM-as-judge validation, sharing the model's blind spots.

We present PRONOIA, an event-driven financial reasoning framework deployed as an open-source web platform with three pillars: (i) an *Event Reasoning Chain* (ERC) architecture formalizing events as state transitions and executing multi-path propagation over an encoded *Propagation Graph*; (ii) a *Predict–Assert–Verify* (PAV) output contract decomposing each event prediction into 3–6 verifiable claims over measurable financial indicators; and (iii) *Reinforcement Learning with Verifiable Rewards* (RLVR) settling every claim against actual post-event market data.

Two controlled experiments on 1,034 time-split out-of-sample A-share events (2024–2026): rule-driven backtest achieves 70.3% direction accuracy with market confirmation as the strongest meta-signal (+18.7pp); PAV-RLVR LoRA adapter improves claim-level accuracy from 57.3% to 63.2% (+5.9pp, paired bootstrap $p < 0.01$) with simultaneous Brier (0.283→0.254) and ECE (0.163→0.107) gains. A controlled ablation shows residual bias correction rewards cause reward hacking, reducing OOS accuracy by 2.29pt—an empirical caution for financial RL design.

Structural analysis reveals asymmetric capability: the model excels at falsifiable claims (FALSE-side 70.9%, significance family 79.1%, horizon $t30$ 75.3%) but struggles with momentum-continuation claims (TRUE-side 53.0%, bm_ret_t15 49.8%). Three case studies characterize failure modes: prior-driven momentum bias, risk paragraphs that are cognitively correct but behaviorally inert, and upstream context contamination.

CCS Concepts: Computing methodologies → Reinforcement learning; Information systems → Expert systems; Applied computing → Economics.

Keywords: Event-driven financial reasoning; LLM agents; Financial world models; RLVR; Event study methodology; Web-based AI agents

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Conference'17, July 2017, Washington, DC, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

ACM Reference Format:
Anonymous Author(s). 2026. PRONOIA: Event-Driven Financial Reasoning with Market-Verifiable Rewards. In . ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

1.1 Relevance to the Web

Financial event analysis is fundamentally a Web problem. Announcements, filings, news, and market data are published on the Web; reasoning agents must retrieve, parse, and synthesize heterogeneous Web sources in real time. PRONOIA is deployed as an open-source Web application[1] where researchers interact through a conversational interface, the system orchestrates multi-agent reasoning over Web-scale financial data, and results are persisted as shareable research assets. The platform's Arena evaluation system enables Web-based collaborative benchmarking of reasoning strategies. This work sits at the intersection of Web-based AI agents, retrieval-augmented reasoning, and online market feedback loops—core themes of the Web conference.

1.2 The Paradigm Shift

Since Ball & Brown [1] and Fama et al. [2], the event study has been the cornerstone of empirical finance. Its five-step pipeline—define event, estimation window, market model, compute AR/CAR, statistical test—has remained structurally unchanged for over half a century. Meanwhile, event-driven trading has evolved from *speed alpha* (co-location, low-latency feeds) to *cognitive alpha*: returns come from reading *what the announcement means* more accurately than the market, not from reading it faster.

LLMs are the natural technology for cognitive alpha—they parse full announcements, detect nuanced guidance tone changes, and sketch transmission chains across sectors. But three fundamental problems block reliable event-driven financial AI, and no existing system solves all three:

Problem 1: Traditional methods describe the past, not the future. Event studies answer "how did *this type* of event move prices on average?" but cannot answer "how will *this specific* event move prices?" Announcements are collapsed into 0/1 dummies; magnitude, wording, and market-surprise component—what actually moves prices [3]—are lost. Macro events are sparse and heterogeneous, limiting out-of-sample predictive power.

Problem 2: Events are treated in isolation, ignoring economic transmission. The standard event study assumes independence [4], but markets react through multi-layer channels: "rate cut → real estate financing improves → building material demand rises → commodity prices move." Markets systematically underreact to indirect effects [5], and the event study pipeline has no slot for them.

[1] https://github.com/vixuowis/Pronoia

Problem 3: LLM-enhanced methods lack external ground truth. LLM event analyses solve Problems 1–2 but validate almost universally with LLM-as-judge—evaluator and evaluatee share error modes, are vulnerable to prompt injection, and "sounding reasonable" ≠ "actually correct."

1.3 The PRONOIA Proposition

PRONOIA resolves this dilemma by grounding every prediction in actual post-event market data:

(1) **Structured Events:** Each event is a structured information shock $(type, direction, strength, surprise)$—not an isolated 0/1 signal—input to multi-path propagation.
(2) **Propagation Graph Reasoning:** Economic logic is explicitly encoded as a DAG $G = (V, A)$ with nodes as macro variables/sectors/assets and edges as pre-built transmission mechanisms (interest-rate channel, supply-demand, industry upstream-downstream, capital flow). Multi-path *search and assembly* replaces free-generation.
(3) **Market Ground Truth Verification:** Every prediction is a falsifiable claim settled by actual post-event market CAR data—not LLM self-assessment. Graph edge weights are continuously calibrated by market feedback, closing the "reason → verify → revise" loop.

1.4 Contributions

(1) **ERC Architecture:** A four-layer decoupled architecture (Perception → Reasoning → Decision → Evolution) with the Event–Transmission–Position (ETP) triple as atomic logic unit, formalizing the Propagation Graph as the financial analog of a robotics world model's transition function.
(2) **PAV-RLVR Framework:** A verifiable output contract (Predict–Assert–Verify) decomposing predictions into 3–6 claim-level assertions over measurable indicators (CAR, benchmark returns, p-values, window averages across T+3/7/15/30/60), combined with RLVR settling every claim against objective market data.
(3) **Controlled Empirical Validation:** (a) Rule-driven backtest isolating the reasoning skeleton: 70.3% ACC with market confirmation (+18.7pp); (b) PAV-RLVR LoRA adapter improving claim accuracy 57.3%→63.2% (+5.9pp, paired bootstrap $p < 0.01$) with Brier and ECE gains.
(4) **Reward Hacking Documentation:** Controlled ablation showing residual bias corrections cause reward hacking (−2.29pt OOS) despite training reward curve improvements—an empirical caution for financial RL design.

2 RELATED WORK

2.1 Event Study Methodology

Foundations by Ball & Brown [1] and Fama et al. [2] established the event study paradigm. Critical limitations documented: event dummies lose surprise magnitude [3]; macro event sparsity limits extrapolation; cross-sectional CAR correlation inflates statistics [4]. We complement this with the Propagation Graph as forward-looking structure preserving event richness and modeling transmission.

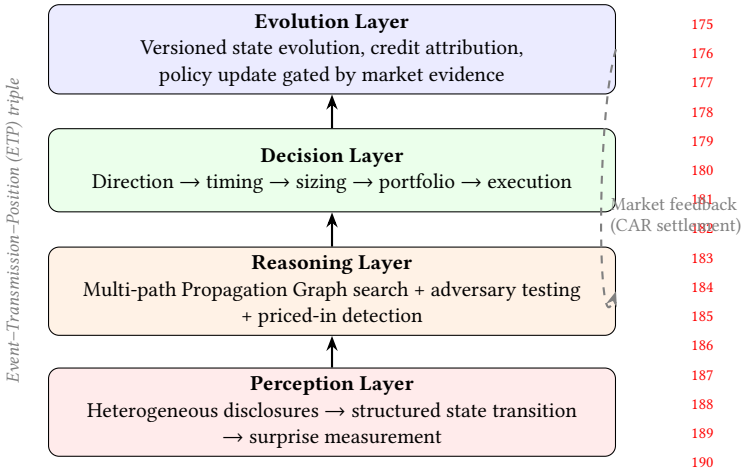


Figure 1: PRONOIA's four-layer Event Reasoning Chain (ERC) architecture. The system operates on the Event–Transmission–Position (ETP) triple as atomic logic unit, with market feedback closing the evolution loop.

2.2 Financial NLP and Event Extraction

The 4th generation of event processing brings LLM-based understanding. MAVEN [8] and MAVEN-ERE [9] provide event detection/coreference datasets; Doc2EDAG [10] and DCFEE [11] tackle document-level financial extraction; FinKario [12] builds event-enhanced financial KGs. These are natural Perception-layer inputs; PRONOIA's contribution is the Reasoning layer—structured propagation over an encoded economic graph with market-grounded RLVR training.

2.3 Web Agents and Tool Use

Recent work on Web agents demonstrates LLMs' ability to navigate complex Web environments [30, 31]. VisualWebArena [32] extends this to multimodal understanding. However, these agents operate in simulated or static Web environments without real-world feedback loops. PRONOIA bridges this gap: the agent reasons over Web-scale financial data (akshare/yfinance APIs, 85+ real data skills) and receives *actual market feedback* as reward signals, creating a live Web-based learning loop.

Retrieval-augmented generation (RAG) [33] grounds LLM outputs in retrieved documents. PRONOIA extends RAG in two directions: (i) retrieval is structured—the Propagation Graph encodes economic mechanisms, not just document similarity; (ii) generation is verifiable—every claim is settled against objective market data, not just grounded in retrieved text.

Tool use benchmarks [34, 35] evaluate LLMs' ability to call external APIs. PRONOIA's 85+ data skills represent a production-scale tool use deployment, but with a critical difference: the system learns *when* and *how* to use tools through RLVR, not just *whether* the tool call succeeds.

2.4 Event-Driven Financial Prediction

Table 1 situates PRONOIA: all prior systems operate on single-hop "event → reaction" mapping with LLM-as-judge or non-verifiable

Table 1: Comparison with event-driven financial AI systems. ACC = direction accuracy on respective test sets.

System	Approach	Limitation	ACC
FinHEAR [13]	Historical analog	Single-chain, no multi-path	58–6
FinKario [12]	KG retrieval	Static KG, retrieval-only	58.1%
TradingAgents [14]	Multi-agent vote	Not event-driven	~55%
Janus-Q [15]	Hierarchical reward	No event modeling	58.7
Argus [16]	LLM-as-judge	Judge shares errors	—
PRONOIA	Graph + PAV + RLVR	US OOS limited	**63.2%**

rewards. The 58–62% ACC ceiling confirms single-hop's limited headroom. PRONOIA's Propagation Graph + RLVR addresses both gaps.

2.5 RLVR and Financial RL

RLVR [17] settles rewards against formal verification rather than self-evaluation. AlphaAgent [18] and FactorMiner [19] apply related ideas to alpha mining. The critical difference: PAV decomposes predictions into claim-level assertions over measurable indicators, making every reward component independently verifiable *and* preventing reward hacking—our ablation in §5.3 documents what happens when this guardrail weakens.

2.6 Robotics World Models

The Propagation Graph formalization draws from robotics world models [6, 7]: $S_{t+1} = f(S_t, a_t)$ maps to our graph: state → market state (sector CAR, volume anomaly, volatility, sentiment), action → structured event, transition function → Propagation Graph (multi-path probability propagation), terminal verification → market CAR direction and claim settlement, planning → Navigator search.

3 SYSTEM DESCRIPTION

3.1 Deployed Platform

PRONOIA is deployed as an open-source Web application (Python 3.10+ backend, React 18 frontend) with the following production components:

- **Conversational Research Workbench:** Researchers interact through a chat interface; cases are persisted in SQLite with full history, enabling refresh-resistant research sessions and follow-up questioning.
- **85+ Real Data Skills:** Market data (akshare/yfinance), news, announcements, financials, macro indicators, and event study methods (AR/CAR computation)—zero mock data.
- **Research Team Mode:** Planner decomposes deep questions → multiple expert agents parallel execution → synthesis → reviewer cross-checks "data facts vs. model inference."
- **Artifact System:** Tool results auto-generate charts, data tables, evidence cards, and one-click research reports.
- **Event-Driven Backtesting Platform:** Real events + Oracle T+3~T+60 CAR labels, Strict/Lenient dual-metric ACC, Wilson CI.

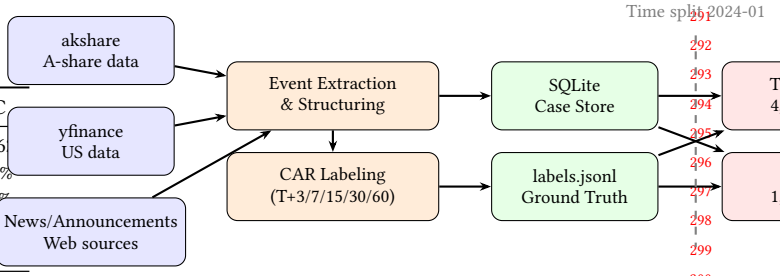


Figure 2: PRONOIA data pipeline: Web data sources → event extraction and CAR labeling → time-split storage for training and evaluation.

- **Arena Horizontal Evaluation:** Multi-run comparison on same dataset—ranking, radar charts, significance tests, head-to-head analysis, cost-effectiveness Pareto frontier.
- **Security:** LLM keys stored only in backend `.env`, never transmitted to browser.

The platform has been iteratively developed through 13 versions (v3.0–v3.13) over 6 months, with the Arena evaluation system and multi-horizon CAR labeling added in recent versions based on experimental feedback.

3.2 Arena Evaluation System

The Arena system enables rigorous horizontal comparison of reasoning strategies on identical datasets:

- **Multi-Run Comparison:** Same dataset, multiple runs—ranking, radar charts, significance tests, event-level head-to-head analysis.
- **Cost-Effectiveness Analysis:** Two-dimensional scatter plots with Pareto frontier (non-dominated sorting + L-shaped reference line), X/Y axis switchable.
- **Pluggable Metrics:** 18+ metrics organized in core/extended/breakdown three-tier system, dynamic column switching on list pages.
- **Statistical Rigor:** Wilson confidence intervals, paired bootstrap tests, FDR correction for multiple comparisons.

This addresses a critical gap in LLM evaluation: most systems report single-run results without statistical validation. Arena makes it trivial to verify whether an improvement is significant or noise.

3.3 Data Pipeline

Figure 2 shows the data pipeline. Events are extracted from Web sources, structured into the ETP format, and labeled with multi-horizon CAR ground truth. The time-split ensures no lookahead bias: training uses events before 2024-01, OOS evaluation uses events after.

4 METHOD

4.1 Financial Event Reasoning Chain

PRONOIA's four-layer architecture (Figure 1) operates on the Event–Transmission–Position (ETP) triple:

(1) **Perception:** Heterogeneous disclosures → structured state transition → surprise measurement.

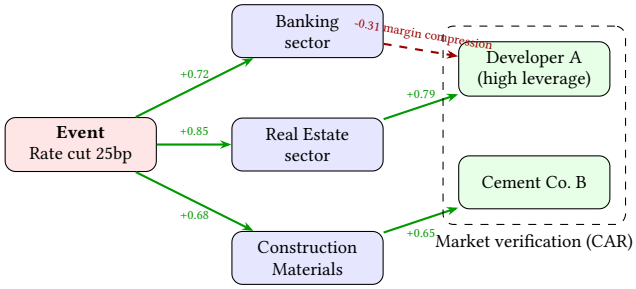


Figure 3: Example Propagation Graph for a rate cut event. Nodes represent sectors/assets with expected direction and magnitude; edges represent transmission mechanisms with conditional probabilities. Dashed red edge shows a competing negative path (margin compression). The final positions are verified against actual post-event CAR data.

(2) **Reasoning:** Multi-path Propagation Graph search + adversary testing + priced-in detection.
(3) **Decision:** Direction → timing → sizing → portfolio → execution.
(4) **Evolution:** Versioned state evolution, credit attribution, policy update gated by market evidence.

This paper focuses on Layers 2 (Reasoning) and 4 (Evolution via RLVR), with Layers 1 and 3 providing input context.

4.2 Events as State Transitions

Definition 4.1 (Event). An event e is a state transition $X_e^- \to X_e^+$ of a declared system S at time t_e, with ex-ante benchmark B_e. The *surprise* is $X_e^+ - B_e$, not $X_e^+ - X_e^-$.

This resolves the common confusion: a large state change (guidance revision \$10–11B → \$8–9B) is not a large surprise if the market expected \$8.5B.

Definition 4.2 (Event Boundary). Boundary = System(S) + State Representation(X) + Analytical Scale(σ). A single earnings release can be one event at filing scale or several at item scale (revenue, EPS, guidance changes).

4.3 Propagation Graph

$G = (V, A)$ is a DAG where:

- **Nodes** V**:** Each v contains (subject, expected_direction, expected_magnitude, time_window), from macro variables to sectors to individual assets.
- **Edges** A**:** Each $a = (u, v, label)$, $label \in \{+1, -1\}$, with conditional transmission probability $P(B_{up}|A_{up}) = N(A_{up} \wedge B_{up})/N(A_{up})$, fused via Bayesian update $P_{new} = P_{prior} \times 0.7 + P_{observed} \times 0.3$.

Three edge-source tiers: (a) prior historical statistics, (b) online recent CAR observations, (c) Navigator real-time reasoning.

Figure 3 illustrates a concrete propagation graph. The rate cut event transmits through multiple paths: direct banking sector impact, real estate financing improvement, and construction materials demand. Each edge weight is continuously calibrated by market feedback.

4.4 Navigator-Searcher Protocol

Six-step collaboration over G:

(1) Navigator decomposes event e into sub-problems, dispatching one Searcher per sub-problem.
(2) Each Searcher, adopting a market-participant persona (long/short/arbitrage) traces one propagation chain.
(3) Outputs merged into DAG: deduplicated, cross-referenced, marked with verification status.
(4) Navigator identifies unconnected components; prompts additional Searchers.
(5) Market verification via CAR + dose-response + volume/volatility validates/falsifies each path.
(6) Verified paths form final reasoning graph; emit directional, magnitude, and confidence judgments.

Key choices: DAG not tree (multiple causes per effect); market posterior not LLM voting as ground truth; game-theoretic branching with heterogeneous personas.

4.5 PAV Output Contract

Predict: Input: event structured fields + as_of research context (strictly no post-event data: T0 return, pre-5/pre-20 drift, CAR to T0) + measurable indicator panel (30 indicators defined, values withheld).

Assert: Model proposes 3–6 verifiable claims: (indicator, horizon, threshold, TRUE/FALSE, confidence, reasoning_chain). Five families across T+3/7/15/30/60: `car_t{h}`, `ret_t{h}`, `bm_ret_t{h}`, `car_t{h}_pvalue`, `car_avg_*`.

Verify: Each claim objectively settled against labels.jsonl. Compute claim-level accuracy, Brier, ECE, verifiability, diversity.

The departure: PAV shifts core metric from "did you guess direction right?" to "are your claims testable and correct?" Evaluation goes from event-level to claim-level.

4.6 RLVR Training

Qwen3-8B (bf16, backbone frozen, LoRA rank 16) trained with GRPO [20] via RLVR [17]. Per event: $G = 4$ candidates, parsed into claims, settled against labels.

Reward is six-element tuple (R0–R5): claim accuracy (core), confidence calibration (Brier), verifiability ratio, diversity (family coverage), format compliance, length penalty.

Ablation (v6.1-b): Adds five residual bias corrections (B3: hard-negative targeting; B4/B5: distribution residual correction; B7: claim combination quality; T3: format bonus). Vanilla (v6.2) disables these.

Training: GRPO (TRL 0.17) + Unsloth acceleration (in-process vLLM, weight sharing, Standby mode, ~2.3× speedup). Batch 8/device × grad-accum 2 = 16 completions/step (4 events × 4 candidates). LR 1e-5, cosine annealing. 1 epoch = 1,241 steps ≈ 4,965 events. Seed 42. Single RTX 4090 (48GB), 10–12h/condition.

5 EXPERIMENTS

5.1 Dataset and Protocol

Known biases: US near-absent in OOS; no macro events in OOS; scope limited to CN A-shares.

Integrity protocol: (1) Strict as_of: features use only pre-event data, audited; (2) Time split only, OOS once; (3) Event-level paired

Table 2: Dataset statistics (time-split).

Split	Events	Markets	Sources
Train	4,965	CN 4,352 / US 613	A-share pre-announcements, M&A, guidance, macro
OOS	1,034	CN 1,027 / US 7	Same, post-training window (2024–2026), one

Table 3: Rule-driven backtest results. S1–S5 implement prior → market confirmation → noise removal → strong-signal focus.

Strategy	N	ACC	95% CI	Sharpe	avgCAR
S1 All	444	51.6%	[47, 56]	−1.01	−0.26%
S2 SectorConfirm	212	**70.3%**	[64, 76]	3.21	+0.71%
S3 StrongEvent	159	71.1%	[64, 78]	1.48	+0.32%
S4 NonPriceIn	130	74.6%	[67, 82]	3.85	+0.65%
S5 StrongNonPriceIn	85	**78.8%**	[69, 87]	0.81	+0.13%

Table 4: PAV-RLVR main results (1,034 OOS events, $t = 0.6$).

Metric	Base	v6.1-b	v6.2-vanilla
Claim ACC	0.5729	0.6093	**0.6322**
Brier	0.2826	0.2569	**0.2539**
ECE	0.1632	**0.1064**	0.1070
Format	0.775	**0.997**	0.842
Settled claims	2,978	4,056	3,154

bootstrap (2,000 resamples, weighted); (4) Claim accuracy weighted by n_settleable; (5) Calibration: Brier + 10-bin ECE + confidence-bin actual accuracy.

5.2 Rule-Driven Backtest (Skeleton Validation)

Design: 27 A-share blue chips + 8 sector ETFs, CSI300 benchmark, 2023-01 to 2026-07, 33 events, 20 types, CAR via market model (120d estimation, 20d gap, T+3 holding).

Key findings: (1) S1 51.6% ≈ random—priors alone cannot trade; (2) Sector ETF confirmation is the strongest meta-signal (+18.7pp, Sharpe −1.01→3.21)—"use market posterior to filter priors"; (3) Event strength quantifies surprise (routine cuts 47.9% < random; 924-package cuts 82.6%); (4) "Graph can only be used once"—low-ACC events (consumer boost 33.3%, spike-crash 31.6%, rate cuts 47.9%) prove lookup tables fail, motivating online reasoning. In/out-of-sample gap 1.2pp; LOEO 68.0–71.4%.

5.3 PAV-RLVR Controlled Ablation

Paired bootstrap: Δ(vanilla − base) = +5.93pt (95% CI [+3.79, +8.07]); Δ(vanilla − v6.1-b) = +2.29pt (95% CI [+0.78, +3.79])—CIs exclude zero. Event-level: vanilla wins 394, v6.1-b 335, tie 302.

Reward hacking: Corrections' only benefit is format (+15.5pt), recoverable by inference-time re-sampling. Training reward (v6.1-b 0.90 vs vanilla 0.69) *inversely correlates* with OOS accuracy (0.609 vs 0.632)—model optimizes proxy signals over claim quality. KL, length, truncation are healthy and comparable—ruling out instability. **Implication: training reward trend cannot validate reward components; OOS adjudication is the only mechanism.**

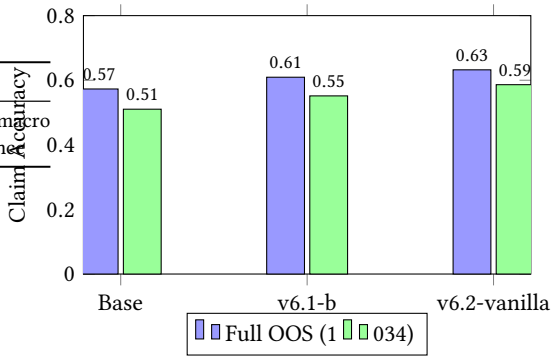


Figure 4: Claim accuracy comparison: full OOS set vs. hard-negative subset. v6.2-vanilla achieves best performance on both, with larger margin on hard negatives (+7.6pt vs. base).

Table 5: Accuracy by event type (CN market).

Event Type	Events	Base	v6.2	Δ
Earnings beat/miss	478	0.524	0.598	**+7.4pt**
M&A/spinoff/financing	373	0.618	0.661	+4.4pt
Guidance up/down	169	0.632	0.679	+4.8pt
Policy rate	7	0.598	0.591	(few)

Table 6: Accuracy by indicator family and horizon.

Slice	Claims	Base	v6.2	Δ
Significance (p-val)	944	0.614	**0.791**	**+17.7pt**
CAR	1,207	0.566	0.592	+2.6pt
Benchmark	647	0.411	0.498	+8.7pt
Horizon t30	675	0.622	**0.753**	**+13.1pt**
Horizon t7	1,154	0.559	0.619	+6.0pt
Horizon t15 (worst)	610	0.452	0.505	+5.3pt

5.4 Hard-Negative Subset

146 events (M&A success/turnaround/beat/additional issuance/buyback): base 0.5103, v6.1-b 0.5515, **vanilla 0.5862**. Corrections help +3.5pt on this subset but net negative overall—the "residual bias correctable by prior" premise is unsupported.

6 ANALYSIS

6.1 By Event Type

Earnings: largest relative lift (+7.4pt) but lowest absolute (0.598)—strongest priors, hardest cases. M&A/guidance: highest absolute (0.661, 0.679).

6.2 By Indicator Family × Horizon

Significance family strongest (+17.7pt to 79.1%): most A-share CARs insignificant—falsifying significance is the model's strongest move. Weakest: `bm_ret_t15` (0.498, coin-flip). Long-horizon t30 (+13.1pt) vs t15 (+5.3pt): falsifiable long-horizon claims outperform.

6.3 Conditional Analysis

Claim count: 2–3 best (0.690/0.656); 4 drops to 0.569. Constraint should narrow to 3–4. **Direction:** FALSE-side **0.709** (n=1,812) vs TRUE-side **0.530** (n=1,342). Training reduces TRUE ratio 0.525→0.425—beneficial RL product matching task asymmetry. **Extremes:** 293 all-correct (28.3%, M&A/earnings); 55 all-wrong (5.3%, beat/turnaround). **Calibration:** Non-monotonic—0.75–0.85 bin actual accuracy only 0.53–0.59 (main ECE contributor); 0.80/0.90 ends reliable (0.777/0.926). Platt post-hoc: ECE 0.085→0.036.

Profile: Excels at FALSE, long-horizon t30, significance, M&A/guidance, 2–3 claims, confidence 0.80/0.90. Struggles with TRUE momentum, benchmark direction, hard-negatives, 4+ claims, 0.75–0.85 confidence.

7 CASE STUDY

7.1 Case 1 (Success): Shengtong Turnaround — vanilla 3/3, base 1/3

002599, 2026-01-30, turnaround. Pre-event CAR −7.95% (oversold); context: "mean-reversion vs micro-profit cancel → neutral, confidence lowered." v6.2 claims: `car_t7>0` TRUE (0.72) ✓; `bm_ret_t30>0` FALSE (0.65) ✓; `car_t30_pvalue<0.05` FALSE (0.55) ✓. Base ignores context, bets p-value TRUE with arbitrary thresholds. **Success = context absorption + diversity planning + betting FALSE.**

7.2 Case 2 (Failure): Zhongjin Gold Beat — all models wrong

600489, 2026-01-28, beat. T0 limit-up +9.98%, pre20 +46% gain. Context warns sell-the-news (−6.0 signal) but all three models claim `car_t7>0` (confidence 0.72), 0/4 wrong. Ground truth: car_t30 −29.8%. **Risk paragraph cognitively correct, behaviorally inert.**

7.3 Case 3 (Failure): Zhongshan Utilities Beat — upstream context wrong

000685, 2026-01-31, beat. Context concludes "volume-price resonance → up"; actual continuous decline (car_t7 −3.2%, car_t30 −17.3%). Model extrapolates weak stock → weak market (bm_ret wrong). pre20 CAR +8.06% used as significance support, not mean-reversion warning. **Garbage-in upstream; features reinforce priors, don't test them.**

7.4 Cross-Case Synthesis

(1) Failures concentrate in "bullish text + T0 positive reaction": prior = momentum, A-share reality = mean reversion.
(2) Risk paragraphs "cognitively correct, behaviorally inert": no hedging claims—prompt/reward design fix.
(3) Benchmark family ≈ coin flip: individual-stock weakness extrapolated to market weakness.
(4) Upstream context with strong priors contaminates downstream—context should output self-confidence + counterparty evidence.
(5) Cross-indicator claims silently dropped—wastes expression, underestimates coverage.

Table 7: Conclusion credibility grading.

Conclusion	Grade	Basis
RLVR effective (+5.9pt)	High	Effect ≫ noise; same pipeline/OOS
Structure (FALSE>TRUE; t30>t15)	High	Hundreds/thousands per cell; 3 models a
Vanilla > v6.1-b 2.29pt	**Low**	3 confounders; CI lower bound +0.78
Confidence 0.90 → 0.926	Verify	Self-selection; n(0.90) unconfirmed
US conclusions	Unusable	OOS US = 7 events

8 DISCUSSION

8.1 Unified Interpretation

Two experiments validate one proposition at three levels:

- **Priors can't trade:** S1 51.6% + base 0.573 confirm priors need purification.
- **Market confirmation purifies:** Rule-side +18.7pp and RL-side +5.9pt are the same mechanism.
- **Environment requires online reasoning:** "Graph once" motivates model-based learning; reward hacking shows even reward functions need OOS adjudication.

8.2 Limitations

Single seed; US OOS = 7 events; 33 rule events + 4,965 RL events insufficient for larger structures; single sampling; TRUE-side weakness in A-shares (short mechanisms incomplete).

8.3 Validity Audit

Recommendation: "Corrections harmful" → "corrections did not produce expected benefits." Supporting the former requires 3-seed × 2-condition matrix.

Three methodological risks: (1) Single-sampling variance unquantified—recommend 5× on 200 events; (2) Multiple comparisons—FDR/Bonferroni + holdout needed for trading filters; (3) as_of strictness unaudited—single risk that could overturn all conclusions.

9 CONCLUSION AND FUTURE WORK

9.1 Takeaways

(1) **Methodology:** Financial event reasoning chain = Propagation Graph + PAV + market posterior reward. Rule backtest (S2 70.3%) and RL ablation (vanilla 0.632 vs base 0.573) validate at two levels.
(2) **Negative results:** Residual corrections reduce OOS accuracy 2.29pt (hard-negatives +3.47pt). Training reward ≠ OOS performance—reward hacking is real in financial RL.
(3) **Structural portrait:** Model excels at falsifiable claims (FALSE 0.709, significance 79.1%, t30 75.3%), struggles with momentum (TRUE 0.530, bm_ret_t15 0.498)—guides prompt/reward design.

9.2 Immediate Next Steps

P0: Hard-negative oversampling (2–3×) + retrain; inference-time format fallback re-sampling. **P1:** Hedging-claim mechanism; pre-drift feature explicit in prompt; US-stratified OOS. **P2:** Claim count narrow to 3–4; parser cross-indicator support; benchmark-family examples. **P3:** Multi-seed validation; Platt calibration retention.

9.3 Mid-Term Evolution

Frequency × horizon 2D reward; scale-conditioned MoE routing (event type × horizon boundaries); data expansion to 500+ rule events + US/macro RL samples.

ACKNOWLEDGMENTS

Anonymous.

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A REPRODUCIBILITY

A.1 Code and Data

PRONOIA is open-source under MIT license:

- **Repository:** `github.com/vixuowis/Pronoia` (Python 3.10+, React 18)
- **Training data:** 4,965 events (akshare/yfinance), structured ETP format
- **OOS data:** 1,034 events, labels.jsonl with all 30 indicator ground truths
- **Backtest data:** 33 events, 27 blue chips + 8 sector ETFs, CSI300 benchmark

A.2 Training Configuration

Table 8: RLVR training hyperparameters.

Parameter	Value
Base model	Qwen3-8B (bf16)
Adapter	LoRA rank 16, α=32, dropout 0.05
Optimizer	AdamW, LR 1e-5, cosine annealing
Batch size	8/device × grad-accum 2 = 16 completions/step
Sampling	G=4 candidates/event, temperature 0.7
Epochs	1 (1,241 steps ≈ 4,965 events)
Hardware	Single RTX 4090 (48GB)
Training time	10–12 hours/condition
Seed	42

A.3 Evaluation Protocol

(1) **Time-split:** Train on events before 2024-01, evaluate on events after (no shuffling).
(2) **Single OOS run:** Each model evaluated exactly once on OOS set (no cherry-picking).
(3) **As-of audit:** All features use only pre-event data; audited for lookahead bias.
(4) **Paired bootstrap:** 2,000 event-level resamples, weighted by n_settleable.
(5) **Calibration:** Brier score + 10-bin ECE + confidence-bin actual accuracy.

A.4 Statistical Tests

All comparisons use paired bootstrap (event-level resampling) with 2,000 iterations. Confidence intervals are 95% percentile intervals. Multiple comparisons (where applicable) use FDR correction at α=0.05. Effect sizes reported alongside p-values.

A.5 Ethics Statement

- All data from public sources (akshare, yfinance); anonymized stock market data.
- No live trading based on this research; all evaluation on historical data.

- Full transparency: reward hacking, overconfidence bias, systematic failure modes documented as negative results for research community, not hidden.
- US events comprise only 7 of 1,034 OOS events; all conclusions limited to A-share market.
- LLM API keys stored only in backend `.env`, never transmitted to browser or logged.